Two Number Sum

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The problem

- Given distinct integers and a target, return any pair summing to the target (empty array if none).

Intuition

- For each number x, we need to know if its partner, target minus x, has already appeared.
- A hash set answers that instantly.

Approach

- 1. Walk the array once
- 2. Check whether target minus x is in the set
- 3. If so, return the pair
- 4. Otherwise add x to the set

Java solution

```java
public class TwoNumberSum {
    public static int[] twoNumberSum(int[] array, int target) {
        Set<Integer> seen = new HashSet<>();
        for (int x : array) {
            if (seen.contains(target - x)) return new int[]{target - x, x};
            seen.add(x);
        }
        return new int[0];
    }
}
```

Complexity

- Time: O(n)
- Space: O(n) using a hash set of seen values

Recap

- For each number x, we need to know if its partner, target minus x, has already appeared.
- Complexity: O(n)